

P YATHIN MACHAIAH

Software & Security Engineer · Distributed Systems · Python Developer

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github.com/PYatiM | leetcode.com/PYatiM | hackerrank.com/PYatiM

EDUCATION

Dayananda Sagar University

B.Tech — Computer Science & Engineering (Cyber Security)

Expected Sep 2027

GPA: 8.38 / 10.0

DSA & Problem Solving: 160+ LeetCode problems (Python) | HackerRank Python 5★ Badge

PROJECTS

APIGATE — Production-Grade Secure API Gateway

Python · FastAPI · OAuth2 / JWT · Redis · OpenTelemetry · Prometheus · Docker Compose

- Architected a horizontally scalable, fault-tolerant API gateway with layered security middleware: OAuth2 JWT authentication, OpenAPI request validation, Redis-backed distributed rate limiting, structured audit logging, and TLS — built for high-availability production deployment.
- Applied OOP design principles (single responsibility, separation of concerns) to produce independently testable, swappable middleware modules — auth, rate limiting, validation, and observability each isolated as discrete components.
- Containerized the full distributed stack with Docker Compose (app + Redis); integrated OpenTelemetry distributed tracing and Prometheus metrics for real-time observability — 58 commits, automated test suite, v1.0.0 released.

Secure Containerized Code Execution Sandbox Engine

Python · FastAPI · Docker · Linux / Syscall Filtering · JavaScript · HTML / CSS

- Designed a scalable, fault-isolated execution engine routing untrusted code through ephemeral Docker containers with enforced CPU quotas, memory caps, and network isolation — mitigating RCE, fork bomb, and privilege escalation attacks at the OS level.
- Engineered a dual-Dockerfile architecture separating API server from runtime sandbox; built a 6-stage sanitization pipeline (Input → API → Runner → Limiter → Sanitizer → Response) to minimize exploitable attack surface at every layer.
- Delivered full end-to-end system design — FastAPI backend and HTML/CSS/JS frontend enabling live code submission and secured output display, demonstrating distributed architecture from API gateway to containerized execution runtime.

TECHNICAL SKILLS

Languages & DSA: Python (asyncio, multiprocessing, sockets) — HackerRank 5★ | Java · C · SQL · Bash | OOP · Data Structures · Algorithms

Systems & Backend: FastAPI · REST APIs · JWT/OAuth2 · Microservices · Redis · PostgreSQL · SQLite · Docker · Git | Distributed Systems · Fault Tolerance

Cybersecurity: Wireshark · Nmap · Burp Suite · Splunk SPL · Metasploit · GDB | OWASP Top 10 · MITRE ATT&CK · NIST CSF · STRIDE

AI / ML: scikit-learn · Isolation Forest · Statistical Anomaly Detection · ML Pipelines · LLM Security · Prompt Injection

Observability: OpenTelemetry · Prometheus | Linux (Kali, Ubuntu) · VMware / VirtualBox · TryHackMe

CERTIFICATIONS

AWS Cloud Practitioner Essentials — Amazon Web Services

2025

Advanced Computer Networks — NPTEL / IIT

2025

Ethical Hacker — Cisco Networking Academy

2026

Google Cybersecurity Certificate

2026

CompTIA Security+ — In Progress

Target Q3 2026

ACHIEVEMENTS & CTF / HACKATHONS

CTF Competitions: Competed in multiple national CTFs covering binary exploitation, web application vulnerabilities, network forensics, and reverse engineering — consistently qualifying past initial elimination rounds.

Hackathons: Participated in multiple hackathons focused on security and systems engineering, building rapid prototypes under time constraints — demonstrating ability to handle ambiguous, broadly defined problem statements.